



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____

Seat Number _____

Student Number

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Family Name _____

First Name _____

School of Mathematics & Physics
Semester Two Examinations, 2023
MATH1061 Discrete Mathematics

This paper is for St Lucia Campus students.

Examination Duration: 120 minutes

Planning Time: 10 minutes

Exam Conditions:

- This is a Closed Book examination - no written materials permitted
- No calculators permitted
- During Planning Time - Students are encouraged to review and plan responses to the exam questions
- This examination paper will be released to the Library

Materials Permitted in the Exam Venue:

(No electronic aids are permitted e.g. laptops, phones)

None

Materials to be supplied to Students:

Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.

None

Instructions to Students:

If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.

Answer all questions.

Total 70 marks.

The Formula Sheet is on the last page

For Examiner Use Only

Question	Mark
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[illegible]

Total

1. (6 marks)

- (a) (2 marks) Write down the quotient and remainder when -27 is divided by 6 , according to the quotient-remainder theorem.

Solution: $-27 = 6 \times -5 + 3$. Thus, the remainder is 3 and the quotient is -5 .

- (b) (2 marks) Determine whether the following statement is true or false. Justify your reasoning.

For all positive integers a and b , we have

$$\gcd(a + b, a - b) = \gcd(a, b).$$

Solution: The statement is false. For instance, $\gcd(3, 1) = 1 \neq 2 = \gcd(3 + 1, 3 - 1)$.

- (c) (2 marks) Compute $\gcd(112, 30)$. Show your work.

Solution: Note that $112 = 2^4 \times 7$ and $30 = 2 \times 3 \times 5$. Thus, the only common factor is 2 ; i.e. $\gcd(112, 30) = 2$.

2. (3 marks) Let p and q be statement variables. Are the following statements logically equivalent? Justify your answer.

Statement A: $(p \rightarrow q) \wedge \sim q$

Statement B: $\sim (p \wedge q)$

Solution: The two statements are not equivalent, because they differ when p is true and q is false. In this case, $p \rightarrow q$ is false, thus Statement A is false. On the other hand, $p \wedge q$ is false, meaning that Statement B is true.

3. (8 marks)

- (a) (3 marks) Let n be a positive integer and let K_n be the complete graph with n vertices. How many edges does K_n have? Explain your reasoning.

Solution: There are n vertices, and there is an edge connecting every pair of distinct vertices. Therefore, selecting an edge is the same as selecting an unordered pair of vertices. There are $\binom{n}{2}$ such choices.

- (b) (5 marks) Let G be a graph. Let y be the number of vertices of G that have odd degree. Prove that y is even.

Solution: Suppose, for the sake of contradiction, that y is odd. We claim that in this case $\sum_{v \in G} \deg(v)$ is odd. Indeed, vertices of even degree do not affect the parity of $\sum_{v \in G} \deg(v)$. Therefore, the parity of $\sum_{v \in G} \deg(v)$ is determined by the vertices with odd degree. If there are an odd number of such vertices, then this sum is odd. But this contradicts the Handshake Theorem, which states that this sum is always even.

4. (6 marks) Let $f : \mathbb{N} \rightarrow \mathbb{R}$ be a function. Prove that the image of $\mathbb{N}$ under f is countable.

Solution: For each y in the image of f , choose exactly one element in $\mathbb{N}$ which maps to y . Let A be the subset of $\mathbb{N}$ consisting of these chosen elements. Note that A is countable, because it is a subset of $\mathbb{N}$. Now the restriction of f to A defines a bijection between A and the image of f . Thus, the image of f is also countable.

Alternate solution: Let $g : \mathbb{N} \rightarrow f(\mathbb{N})$ be defined by $g(n) := f(n)$. i.e. g is the same as f but the codomain is restricted. By definition, g is surjective. Therefore $|\mathbb{N}| \geq |f(\mathbb{N})|$.

5. (8 marks) Let $f : \mathbb{Z} \rightarrow \{0, 1, 2, 3, 4\}$ be defined by: $f(n) = n^2 \pmod{5}$.

For example, $f(7) = 4$ because $7^2 = 49$ is congruent to 4 modulo 5.

- (a) (2 marks) What is $f(2023)$? Justify your answer.

Solution: We have $f(2023) = f(3) = 9 \pmod{5} = 4$.

- (b) (2 marks) What is the preimage of 1? Justify your answer

Solution: The preimage is equal to the set integers of the form $5k \pm 1$. Indeed, every such integer is in the preimage because $f(5k \pm 1) = 1$, and no other integer is in the preimage because $f(5k) = 0$ and $f(5k \pm 2) = 4$.

- (c) (2 marks) Is f one to one? Justify your answer.

Solution: f is not one to one because, for instance, $f(1) = f(-1)$, but $1 \neq -1$.

- (d) (2 marks) Is f onto? Justify your answer.

Solution: From (b), we know that f has image $\{0, 1, 4\}$. Therefore f is not onto because 3 is in the codomain of f but not the image.

6. (8 marks) Let $(G, *)$ be a group. Define a relation $\sim$ on G as follows: $a \sim b$ if there exists $g \in G$ such that $g * a * g^{-1} = b$. Prove that $\sim$ is an equivalence relation on G .

Solution: We show that the relation is reflexive, symmetric and transitive.

Reflexivity: Let $a \in G$. We know that $1_G a 1_G^{-1} = a$, where 1_G is the group identity. Therefore $a \sim a$.

Symmetry: Suppose that $a, b \in G$ and that $a \sim b$. Therefore $gag^{-1} = b$ for some $g \in G$. Therefore $g^{-1}bg = a$. This means that $g^{-1}b(g^{-1})^{-1} = a$. Since $g^{-1} \in G$, this means that $b \sim a$.

Transitivity: Suppose that $a, b, c \in G$, $a \sim b$ and $b \sim c$. Then $gag^{-1} = b$ for some $g \in G$, and $hbh^{-1} = c$ for some $h \in G$. Now observe that $(hg)a(hg)^{-1} = h(gag^{-1})h^{-1} = hbh^{-1} = c$. Since $hg \in G$, this means that $a \sim c$.

7. (8 marks) Recall that for a set A , its power set is denoted $\mathcal{P}(A)$.

(a) (4 marks) Compute $\mathcal{P}(\{x\})$ and $\mathcal{P}(\mathcal{P}(\{x\}))$.

Solution: We have

$$\mathcal{P}(\{x\}) = \{\emptyset, \{x\}\}$$

and

$$\mathcal{P}(\mathcal{P}(\{x\})) = \mathcal{P}(\{\emptyset, \{x\}\}) = \{\emptyset, \{\emptyset\}, \{\{x\}\}, \{\emptyset, \{x\}\}\}.$$

(b) (4 marks) Prove that

$$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B).$$

Solution: Indeed,

$$\begin{aligned} X \in \mathcal{P}(A \cap B) &\iff X \subseteq A \cap B \\ &\iff X \subseteq A \text{ \& } X \subseteq B \\ &\iff X \in \mathcal{P}(A) \text{ \& } X \in \mathcal{P}(B) \\ &\iff X \in \mathcal{P}(A) \cap \mathcal{P}(B). \end{aligned}$$

8. (6 marks) Consider the sequence $(a_n)_{n=1}^{\infty}$ defined recursively by $a_1 = 3$, $a_2 = 8$, and

$$a_n = 4(a_{n-1} - a_{n-2}), \quad \forall n \geq 3.$$

Using strong induction, prove that $a_n = 2^{n-1}(n+2)$ for all integers $n \geq 1$.

Solution: Note that

$$a_1 = 3 = 2^0 \cdot 3 = 2^{1-1}(1+2),$$

and

$$a_2 = 8 = 2 \cdot 4 = 2^{2-1}(2+2).$$

Thus, our base cases are covered. Suppose the hypothesis is true for all $k < n$. Then,

$$a_n = 4a_{n-1} - 4a_{n-2} = 4(2^{n-2}(n+1) - 2^{n-3}n) = 4 \cdot 2^{n-3}(2n+2-n) = 2^{n-1}(n+2).$$

9. (7 marks) A group of 10 friends are attending the movies together and insist on sitting in one row. There are exactly ten seats in each row.

- (a) (2 marks) Two of the friends (Bailey and Courtney) insist on sitting next to each other. In how many different ways can the 10 be seated while satisfying this wish? Justify your answer.

Solution: We treat the two people sitting side-by-side as one person. This gives $9!$ ways for the 10 people to sit. But for each of these ways, the two people who insisted on sitting side-by-side can be swapped, still remaining side-by-side and yielding a new way to sit. Therefore there are $2 \cdot 9!$ ways.

- (b) (2 marks) Now suppose Bailey and Courtney no longer insist on sitting next to each other. However, there are two other people, Tara and Adam, who do not like each other and do not want to sit next to each other. In how many different ways can the 10 be seated while satisfying this wish? Justify your answer.

Solution: There are $10!$ ways for the group to sit. Out of these $10!$ ways, $2 \cdot 9!$ ways will have them sitting side-by-side. Therefore there are $10! - 2 \cdot 9!$ ways where they are not sitting side-by-side.

- (c) (3 marks) Finally, suppose Bailey and Courtney want to sit next to each other and, at the same time, Tara and Adam do not want to sit next to each other. In how many different ways can the 10 be seated while satisfying these wishes? Justify your answer.

Solution: We treat Bailey and Courtney as one person. So then we have to sit 9 people 2 of whom don't want to sit next to each other. By the previous part, the total is $9! - 2 \cdot 8!$. But we can also permute the location of Bailey and Courtney; thus, the total is $2(9! - 2 \cdot 8!)$.

10. (10 marks) Define a binary operation $\circ$ on $G = \mathbb{R}^2 - \{(0, 0)\}$ by

$$(a, b) \circ (c, d) = (ac - bd, ad + bc).$$

- (a) (6 marks) Prove that $(G, \circ)$ is an abelian group. (You do not need to prove that the operation is closed or associative.)

Solution: We show the existence of an identity, the existence of inverse, and commutativity.

Identity: We claim $(1, 0)$ is the identity. This follows from

$$(a, b) \circ (1, 0) = (a, b) = (1, 0) \circ (a, b).$$

Inverses: We have

$$(a, b) \circ (a, -b) = (a, -b) \circ (a, b) = (a^2 + b^2, 0).$$

Thus, every element (a, b) has an inverse; namely, $\frac{1}{a^2+b^2}(a, -b)$.

Commutativity: Finally, the group is commutative because

$$(a, b) \circ (c, d) = (ac - bd, ad + bc) = (ca - db, cb + da) = (c, d) \circ (a, b).$$

Alternate solution sketch: Identify $\mathbb{R}^2$ with $\mathbb{C}$ via the map $(a, b) \mapsto a + bi$. Then our set gets identified with $\mathbb{C}^\times$ and the above operation is the natural multiplication on $\mathbb{C}^\times$.

- (b) (4 marks) Determine all subgroups of $(G, \circ)$ which have exactly two elements. Justify your answer.

Solution: We are looking for elements $(a, b) \in G$ that satisfy $(a, b) \circ (a, b) = (1, 0)$. This reduces to $a^2 - b^2 = 1$ and $2ab = 0$. This second equality means that $a = 0$ or $b = 0$, excluding the case that $a = b = 0$ since $(0, 0) \notin G$. If $a = 0$ then $-b^2 = 1$ has no solution in $\mathbb{R}$. On the other hand, if $b = 0$ then $a^2 = 1$ has solutions $a = \pm 1$. Therefore, the desired elements are $(\pm 1, 0)$ and the only subgroup of order two is $\{(1, 0), (-1, 0)\}$.

End of Examination

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Logical Equivalences

Given any statement variables p , q and r , a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of $\mathbf{t}$ and $\mathbf{c}$</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$

Set Identities For all sets A , B and C that are subsets of a universal set U :

1. Commutative Laws (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$
2. Associative Laws (a) $(A \cup B) \cup C = A \cup (B \cup C)$
(b) $(A \cap B) \cap C = A \cap (B \cap C)$
3. Distributive Laws (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity Laws (a) $A \cup \emptyset = A$ (b) $A \cap U = A$
5. Complement Laws (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$
6. Double Complement Law $(A^c)^c = A$
7. Idempotent Laws (a) $A \cup A = A$ (b) $A \cap A = A$
8. Universal Bound Laws (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$
9. De Morgan's Laws (a) $(A \cup B)^c = A^c \cap B^c$ (b) $(A \cap B)^c = A^c \cup B^c$
10. Absorption Laws (a) $A \cup (A \cap B) = A$ (b) $A \cap (A \cup B) = A$
11. Complement of U and $\emptyset$ (a) $U^c = \emptyset$ (b) $\emptyset^c = U$
12. Set Difference Law $A - B = A \cap B^c$

The Quotient-Remainder Theorem Given any integer n and any positive integer d , there exist unique integers q and r such that $n = dq + r$ and $0 \leq r < d$.

Unique Factorisation Theorem Given any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$, and positive integers $e_1, e_2, \dots, e_k$ such that $n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, and any other expression for n as a product of prime numbers is identical to this except perhaps for the order in which the factors are written.

Schröder-Bernstein Theorem For all sets A and B , if $|A| \leq |B|$ and $|A| \geq |B|$ then $|A| = |B|$.

Binomial Theorem Given any real numbers a and b , and any nonnegative integer n ,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.$$